

Pavan Timmanna Gaonkar

India

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Education

Masters in Physics , Indian Institute of Technology Hyderabad	2025–Present
Bachelor's in Physics and Mathematics , Karnataka University, Dharwad CGPA: 8.3	2021–2024
Pre-University Education (12th), YTSS PU College, Yellapur Distinction (535/600)	2019–2021
SSLC (10th), Government High School, Bisgod 98.24% (614/625)	2018–2019

Research Projects

Research Project in Topological Condensed Matter Theory

Implemented the Haldane model through analytical calculations and Python-based numerical simulations, including the computation of Berry curvature and Chern numbers in the discretized Brillouin zone, and investigated topologically protected chiral edge states on cylindrical geometries.

Ongoing Research Project in Computational Materials Physics

Using density functional theory (DFT) and VASP to investigate candidate topological thermoelectric materials through first-principles electronic structure calculations and computational screening of materials.

Computational Physics Project (Group Project under Prof. Anupam Gupta)

Developed numerical simulations of driven damped harmonic oscillators under deterministic and stochastic forcing using Python. Implemented fourth-order Runge–Kutta and Euler–Maruyama methods to solve Langevin dynamics with Ornstein–Uhlenbeck noise, and analyzed phase-space trajectories, energy dissipation, autocorrelation functions, and power spectral density.

Research Interests

I have been fascinated by how the complex phenomena we see around us can be described by a theory, and by the idea of emergence. I enjoyed my statistical mechanics and quantum mechanics courses, and I am working towards many-body physics in condensed matter theory – particularly topological phases of matter, the quantum Hall effect, and quantum geometry, alongside computational materials physics.

Technical Skills

- **Programming:** Python, NumPy, SciPy, Matplotlib
- **Computational:** DFT (VASP)
- **Mathematics:** Linear Algebra, Group Theory, Differential Equations, Fourier Analysis

Other Interests & Languages

- **Interests:** Indian Philosophy; Modern and older Kannada Literature
- **Languages:** English, Kannada, Hindi